

A. Data

Source: Department of Labor and Employment (DOLE) – National Wages and Productivity Commission (NWPC).

URL: <https://nwpc.dole.gov.ph/>

Variables:

- **x (Independent):** Year (e.g., 2008 to 2023)
- **y (Dependent):** Daily Minimum Wage Rate in NCR for Non-Agriculture Workers (in PHP)

Data table

Year (x)	Minimum Wage Rate in PHP (y)
2008	382
2009	382
2010	404
2011	426
2012	446
2013	456
2014	466
2015	481
2016	481
2017	512
2018	537
2019	537
2020	537
2021	537
2022	570
2023	610

B. Phyton Script

```
import matplotlib.pyplot as plt
import numpy as np

# Data
x = np.array(
    [
        2008,
        2009,
        2010,
        2011,
        2012,
        2013,
        2014,
        2015,
        2016,
        2017,
        2018,
        2019,
        2020,
        2021,
        2022,
        2023,
    ],
    dtype=float,
)
y = np.array(
    [
        382,
        382,
        404,
        426,
        446,
        456,
        466,
        481,
        481,
        512,
        537,
        537,
        537,
        537,
    ]
)
```

```

        570,
        610,
    ],
    dtype=float,
)

# Least squares regression parameters
n = len(x)
a1 = (n * np.sum(x * y) - np.sum(x) * np.sum(y)) / (
    n * np.sum(x**2) - (np.sum(x)) ** 2
)
a0 = (np.sum(y) - a1 * np.sum(x)) / n

# Calculate Residuals (e = y - y_pred)
y_pred = a0 + a1 * x
residuals = y - y_pred

# Generate and Save Residual Plot
plt.figure(figsize=(8, 4))
plt.scatter(
    x, residuals, color="#1f4e79", edgecolors="black", s=50, zorder=3
)
plt.axhline(0, color="#d9534f", linestyle="--", linewidth=1.5)
plt.xlabel("Year (x)")
plt.ylabel("Residuals (PHP)")
plt.title("Residual Plot: NCR Minimum Wage Regression")
plt.grid(True, linestyle="--", alpha=0.6)
plt.tight_layout()

# SAVE BEFORE SHOWING
plt.savefig("wage_residual_plot.png", dpi=300)
plt.show()

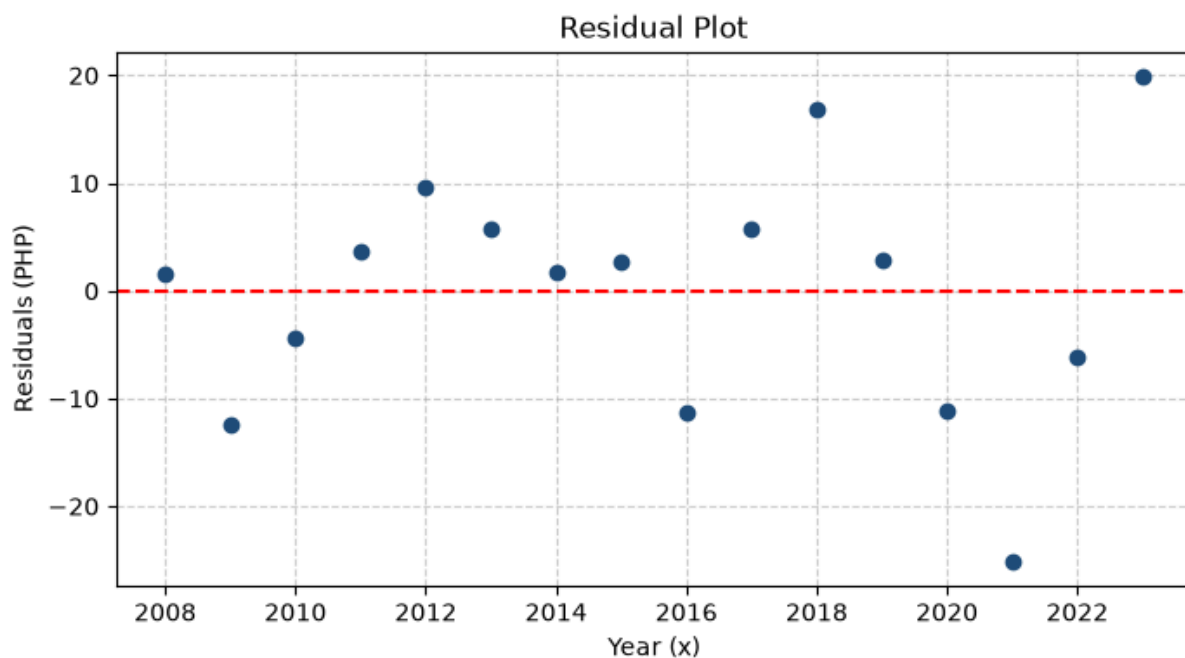
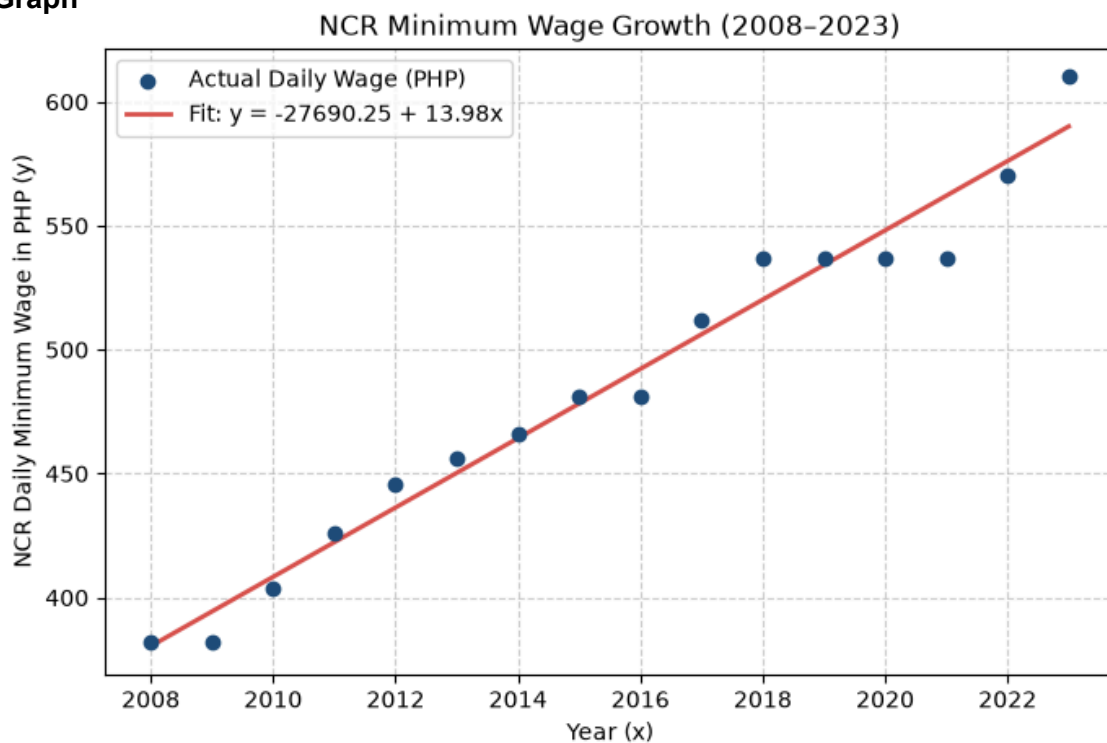
```

C. Regression Results

Calculated Output Summary

- **Regression Equation:** $y = -27690.2544 + 13.9794x$
- **Slope (a_1):** 13.9794 PHP/year
- **Intercept (a_0):** -27690.2544 PHP
- **Sum of Squared Errors (Sr/ SSE):** 1964.8559
- **Coefficient of Determination (r^2):** 0.9713
- **Standard Error ($s_{y/x}$):** 11.8468 PHP

Graph



Prediction Calculation

- **Substitution:** $y = -27690.2544 + 13.9794(2025)$
- **Predicted Result:** PHP 618.05
- **Meaning:** Based on historical wage growth trends, the estimated minimum wage in NCR for 2025 is approximately PHP 618.05 per day.

Interpretations

- **Slope (a_1):** On average, the NCR daily minimum wage increased by approximately PHP 13.98 per year between 2008 and 2023.
- **Intercept (a_0):** The intercept (a_0) is negative because $x = 0$ means the year 0 AD, which doesn't make sense for modern wages. This happens because we used calendar years (like 2008) instead of starting at Year 0.
- **Fit Quality (r^2 and $s_{y/x}$):** An r^2 value of 0.9713 indicates that 97.13% of the total variation in NCR minimum wage is explained by time. The standard error ($s_{y/x}$) shows that predicted values vary from actual wage values by about PHP 11.85 on average.
- **Residual Analysis:** The points bounce above and below zero because minimum wage stayed frozen for a few years (like 2018–2021) and then suddenly jumped up when new wage laws were passed.